

Q-II. Answer precisely in the space provided

[30 min; $4 \times 7.5 = 30$ points]

1. Bag I contain w_1 white and b_1 black balls. Bag II contains w_2 white and b_2 black balls.

A ball is drawn at random from one of the bags, and it is found to be white. What is the probability that it was drawn from Bag I.

Write all the relevant probabilities and finally compute the required one using Bayes theorem. (write it as an expression in terms of w_1 , b_1 , w_2 and b_2)

2. We know that a kernel $\kappa(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^T \phi(\mathbf{q})$. Consider a data set of N samples $\{\mathbf{x}_i\}$ and a Kernel Matrix K such that (i, j) th element of this matrix is $\kappa(\mathbf{x}_i, \mathbf{x}_j)$. (a) Show that K symmetric (b) Show that Kernel matrix (i.e., K) is PSD (Positive Semi Definite)

3. Consider a function $f(w) = w^2 + 2w + 1$. We want to minimize $f(w)$ using gradient descent. Initialize w_0 as 5. Choose the learning rate η as 0.1. (a) Show two iterations i.e., compute the value of w after first and second iterations. i.e., w^1 and w^2 . (b) Suggest a learning rate better than 0.1 so that it converges faster. (c) Suggest a learning rate that will take you to the optimal solution in one step i.e., w^1 .

4. Consider a simple linear regression problem with $x_i \in R$ and $y_i \in R$. We would like to learn a model of $y_i = w_1 x_i + w_2$. Given the data $\mathcal{D}$ of $N = 4$ samples, as $\{(1, 2), (2, 3), (3, 4), (4, 5)\}$. Without numerically computing, guess/find $\mathbf{w} = [w_1, w_2]^T$ by observing the data.

Assume a generic linear regression model of $y_i = \theta^T \mathbf{x}_i + b$, with $\mathbf{x}_i \in R^d$. Find θ and b using either a closed form expression or gradient descent. Write the python snippet to do this clearly showing the matrix, vector manipulations. (do not use any direct api calls for linear regression, but can use numpy).

Q-III. Answer in detail in the space provided

[30 min; $2 \times 15 = 30$ points]

1. Consider 10 samples $\{x_1, \dots, x_{10}\}$ (given below) to be clustered into three clusters using K-Means algorithm. Algorithm is initialized with an assignment as Cluster-1 = $\{x_1, x_4, x_7, x_{10}\}$, Cluster-2 = $\{x_2, x_5, x_8\}$ and Cluster-3 = $\{x_3, x_6, x_9\}$. (i) Write the steps of the K-Means with appropriate equations (ii) Numerically show K-means for 3 iterations (show means and clusters in the three iterations)

$$\left\{ \begin{array}{c} \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] \left[\begin{array}{c} -1 \\ -1 \\ -1 \end{array} \right] \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right] \left[\begin{array}{c} 5 \\ 5 \\ 0 \end{array} \right] \left[\begin{array}{c} 4 \\ 4 \\ 0 \end{array} \right] \left[\begin{array}{c} 6 \\ 6 \\ 0 \end{array} \right] \left[\begin{array}{c} 6 \\ 4 \\ 0 \end{array} \right] \left[\begin{array}{c} 4 \\ 6 \\ 0 \end{array} \right] \left[\begin{array}{c} 0 \\ -6 \\ -4 \end{array} \right] \left[\begin{array}{c} 0 \\ -4 \\ -6 \end{array} \right] \end{array} \right\}$$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7 \quad x_8 \quad x_9 \quad x_{10}$

2. We know PCA (or Eigen Face) can be used for dimensionality reduction (say from 10^4 to 10^2) and then use a simple classifier for tasks like face recognition.

We are given a training set of 100 face examples each of two individuals "A" and "B" (i.e., 200 face images), each of size 100×100 (first 100 are from "A" and second 100 are from "B"). We are given a new "test image" "Q" of size 100×100 . Decide whether this face is of "A" or "B".

Write the precise equations/steps and python snippets (use numpy, if required) for (i) Computing mean and co-variance (ii) Computing and choosing right eigen vectors (iii) Reducing dimensionality of training data from 10^4 to a lower appropriate dimension (say d found in step (ii)) and (iv) Doing a K-NN based classification of the test sample/image "Q" (Assume $K=5$).